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A randomised controlled pilot study of the management of gestational impaired glucose tolerance

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Objective To determine whether a study of a less intensive form of management for impaired glucose tolerance in pregnancy is feasible and whether women would accept randomisation.

Design Prospective randomised controlled study.

Setting A large district general hospital and a large teaching hospital in West Yorkshire.

Sample Seventy women with impaired glucose tolerance in pregnancy.

Methods One group monitored plasma glucose up to four times daily. The other group did not monitor plasma glucose at all.

Main outcome measures The number of women recruited of those approached and neonatal admissions to special care baby units in each group.

Results Sixty-eight of 70 women approached entered the study. There were no statistically significant differences between the groups in neonatal outcome measures. The median number of plasma glucose measurements in the monitored group was 118 (range 0–500), and 19% of women in the monitored group were treated with insulin.

Conclusions This study fails to demonstrate any benefit from intensive management of impaired glucose tolerance in pregnancy with additional maternal inconvenience. This pilot study has shown that a large randomised controlled study of the management of impaired glucose tolerance in pregnancy is not only feasible but necessary.

INTRODUCTION

Abnormal glucose tolerance arising in pregnancy affects between 3% and 6% of all pregnant women and is reported to be associated with adverse pregnancy outcome[1–3]. As a consequence, various intensified management regimens are used for women with gestational abnormalities of glucose tolerance[4–9]. The aim is to return glucose control to normal and improve perinatal outcome. This involves an increased level of maternal inconvenience compared with standard antenatal care, usually in the form of more frequent antenatal visits, daily capillary glucose series, early delivery and possibly an increased caesarean section rate. There is some evidence that management aimed at improving glucose control in pregnancy may improve perinatal outcome in women with frank gestational diabetes mellitus[4,5,8] but not in women with only mild abnormalities of glucose tolerance. Previously the World Health Organisation defined mild abnormalities of glucose control as impaired glucose tolerance when fasting plasma glucose was < 7·8 mmol/L and plasma glucose level 2 hours after the ingestion of 75 g glucose was 7·8 to 11·0 mmol/L. Current World Health Organisation recommendations, however, are to abandon this sub-division and classify all abnormal glucose metabolism arising in pregnancy as gestational diabetes[10].

Only one randomised controlled study of the management of women with gestational impaired glucose tolerance has been undertaken[11]. In this study, women with impaired glucose tolerance were randomly allocated to either standard antenatal care or intensified antenatal care. Intensified care consisted of dietary advice, capillary glucose monitoring, serial ultrasound and cardiotocograph monitoring of fetal wellbeing. The group receiving standard care delivered infants with a statistically significantly higher birthweight but delivery occurred one week later. This is unlikely to be of clinical relevance. There was no difference in adverse perinatal outcomes between the two groups.

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However, most clinicians manage women with gestational impaired glucose tolerance in the same way as those with diabetes. Moreover, many clinicians are reluctant to consider reducing intensive management despite a lack of evidence of benefit. We performed this pilot study to determine whether it would be possible to randomise women to a less intensive regime of care.

METHODS

Women were recruited from two specialist diabetic/antenatal clinics after referral from general antenatal clinics. The inclusion criteria were impaired glucose tolerance as defined by a fasting blood glucose level < 7·0 mmol/L and a blood glucose level 2 hours after the ingestion of 75 g glucose of between 7·8 and 11 mmol/L. Glucose tolerance tests had been performed at the discretion of individual clinicians.

The study was of prospective, randomised, controlled design. Women were randomly assigned to either intensive diabetic monitoring (monitored group) or to no diabetic monitoring (unmonitored group). They gave written informed consent to participation. Randomisation was by computer generated code to one of two management arms, stratified for trimester of diagnosis and ethnicity. Randomisation was controlled from a trial centre and administered by telephone. Ethics committee approval was obtained at both hospitals before commencing the study.

Women recruited into this study were cared for in a combined diabetic antenatal clinic run jointly by a diabetologist and an obstetrician. The diabetologist was aware of the group to which each woman was randomised, while the obstetrician was blinded to minimise the risk of bias in decisions related to obstetric management. This was achieved by the women being seen separately by the two clinicians, who did not discuss the diabetic management of individuals.

The primary outcome measure was frequency of admission to a special care baby unit. Secondary outcome measures included measures of perinatal morbidity, such as birth trauma, metabolic disturbance, gestation at delivery and birthweight. Measures of maternal inconvenience including number of capillary samples performed for glucose estimation, number of antenatal clinic visits and mode of delivery were also analysed.

Women randomised to the monitored group were given standard dietary advice about restricting carbohydrate intake to 185 g/day and a diet sheet listing calorific values of common foods. Their glucose metabolism was monitored by means of capillary glucose series (capillary glucose samples obtained one to two hours after meals) performed five times per week and glycosylated haemoglobin measurements performed monthly. Insulin was introduced if five or more capillary glucose measurements were > 7·0 mmol/L in one week. Antenatal care consisted of routine assessment with serial ultrasound for growth, amniotic fluid and Doppler studies of umbilical artery waveforms. The gestation and mode of delivery were determined by the obstetrician in discussion with the woman, but took place no later than 41 weeks of gestation.

Women randomised to the unmonitored group were given dietary advice, as above, but their glucose metabolism was not monitored. Glycosylated haemoglobin measurements were performed monthly, but the results were not made available during the study. If clinicians were concerned about the possibility of deteriorating glucose metabolism in these women, they could be withdrawn from the study and formally monitored as above. Antenatal care was identical to that offered to the monitored group.

Maternal age, body mass index, glycosylated haemoglobin levels, birthweight and birthweight ratios were normally distributed. Values are presented as mean (SD) and were compared between groups using the Student's t test. Values for plasma glucose during glucose tolerance testing did not conform to a normal distribution and these and other discrete variables are presented as median (range) and were compared between groups using the Mann-Whitney U test. Categorical variables were compared using the two-tailed Fisher's exact test and when there were more than two categories by the Pearson χ^2 test. A P value of < 0·05 was taken to indicate significance. Statistical analyses were performed using SPSS Version 8·0.

RESULTS

Seventy women fulfilled the entry criteria during the study period. Sixty-eight women agreed to enter the study. One woman declined, preferring intensive monitoring, and the other declined because she would not accept monitoring. All the others gave written informed consent to entering the study; 36 were randomly assigned to the unmonitored group and 32 to the monitored group.

The two groups were demographically similar (Table 1). At admission, the two hour plasma glucose level was statistically significantly higher in the unmonitored group (Table 2). Glycosylated haemoglobin levels were significantly higher in women in the unmonitored group at 32 weeks but not at any other stage (Table 3).

There were no statistically significant differences in neonatal outcome (Table 4). There were two congenital abnormalities in the unmonitored group (cystic fibrosis and a small polyp on a finger) and one in the monitored group (congenital heart disease). One shoulder dystocia occurred in the unmonitored group, resulting in admission to a special care baby unit but no long term consequences. There were no stillbirths or neonatal deaths in either group. There were no statistically significant

Table 1. Demographic characteristics of women participating in the study. Values are given as *n* (%), median [range] or mean {SD}. BMI = body mass index.

Characteristics	Monitored group (*n* = 32)	Unmonitored group (*n* = 36)
Age at delivery (years)	29·7 {6·23}	31·9 {5·17}
Weight at booking (kg)	76·4 {17·3}	74·1 {19·8}
BMI at booking	31·2 {6·7}	27·5 {6·1}
Parity	2 [0–6]	1 [0–9]
Asian women	10 (31)	11 (31)
Caucasian women	22 (69)	25 (69)
Family history of type 2 diabetes	12 (37)	11 (31)
Large for dates	10 (31)	11 (31)
Glycosuria	15 (41)	25 (69)

differences between the two groups in gestation at delivery, birthweight, birthweight ratio and frequency of birthweight > 90th centile for gestation. (Table 4).

Maternal outcome measures are shown in Table 5. As expected, women in the unmonitored group performed no capillary glucose samples whereas those in the monitored group performed a median of 118 samples. No women in the unmonitored group were commenced on insulin, whereas six women in the monitored group (19%) required such treatment. There were no differences in the number of antenatal clinic outpatient visits, hospital admissions or in rates of operative delivery (Table 5).

The postnatal glucose metabolism data were consistent with the antenatal findings (Table 6). The two-hour glucose level and the frequency of postnatal diabetes and impaired glucose tolerance tended to be greater in the unmonitored group although the differences between the two groups were not statistically significant.

DISCUSSION

This pilot study has confirmed the feasibility of conducting a multicentre, randomised controlled study of different regimes for the management of abnormal glucose tolerance in pregnancy. Sixty-eight of 70 women who were approached agreed to entry. The other two women opted for a particular management regime, different in each case. Many of the other women in the trial would have preferred not to perform capillary glucose samples. The 'default' option, however, was standard care (i.e. to perform capillary samples).

Previously, the study by Li *et al.*[11] has shown no benefit for intensive management of impaired glucose tolerance in terms of maternal or fetal outcome. Recently, a study[12] of 300 women with gestational diabetes included both women with impaired glucose tolerance and those with frank diabetes mellitus. Strict glycaemic control with tertiary antenatal care was compared with routine antenatal care and again no significant difference in birthweight, macrosomia or neonatal hypoglycaemia was demonstrated. There was no birth trauma in either group.

In the light of these findings it would seem that intensive regimens, particularly in impaired glucose tolerance have been introduced without appropriate evidence. Such regimens have not been shown to decrease perinatal morbidity nor have potential costs to the mother been assessed. This study provides no support for intensive regimens but rather confirms the need for a large trial of such regimens and demonstrates that it is feasible.

The results of this study are difficult to interpret clearly for two reasons. First, the study population is small, primarily because the study was intended as a pilot project. We recognised that to recruit enough women to demonstrate that the trial was feasible in our two centres would take two years. We then intended to look for funding to open out the study to other centres. However, a much larger international randomised controlled trial (ACHOIS*) with a similar protocol has now commenced recruitment, and it was felt more appropriate to contribute to this study rather than duplicate it.

Table 2. Glucose metabolism data at entry into the study. Values are given as median (range) or mean [SD]. HbA1c = glycosylated haemoglobin.

	Monitored group (*n* = 32)	Unmonitored group (*n* = 36)
Gestation at entry (weeks)	31 (24–38)	32 (15–37)
HbA1c (%)	5·3 [0·83]	5·6 [0·96]
Fasting glucose (mmol/L)	4·6 (3·5–5·8)	4·7 (3·5–7·0)
Two-hour glucose (mmol/L)*	8·5 (7·9–10·8)	8·9 (7·8–11·0)

*$P = 0·025$.

Table 3. Glycosylated haemoglobin (HbA1c) during the course of the study. Values are given as mean (SD).

HbA1c (%)	Monitored group	Unmonitored group
28 weeks	4·9 (0·7) (*n* = 8)	5·5 (1·1) (*n* = 8)
32 weeks*	5·2 (0·8) (*n* = 20)	5·9 (1·3) (*n* = 19)
36 weeks	5·3 (0·8) (*n* = 31)	5·6 (1·2) (*n* = 32)
38 weeks	5·3 (0·9) (*n* = 24)	5·5 (0·9) (*n* = 27)
At term	5·1 (0·8) (*n* = 10)	5·5 (0·9) (*n* = 10)

*$P = 0·03$.

*For Further details of the ACHOIS study, contact Caroline Crowther on achois@medicine.adelaide.edu.au.

Table 4. Neonatal outcome measures. Values are given as *n* (%), mean [SD] or median {range}. SCBU = special care baby unit.

	Monitored group (*n* = 32)	Unmonitored group (*n* = 36)
Frequency of admission to SCBU	2 (6)	6 (17)
Frequency of hypoglycaemia	2 (6)	6 (17)
Gestation at delivery (weeks)	39 {36–41}	39 {34–41}
Birthweight (kg)	3·58 [0·55]	3·62 [0·55]
Birthweight ratio	1·07 [0·16]	1·10 [0·15]
Birthweight > 90th centile for gestation	8 [25]	7 [19]

Table 5. Maternal outcome measures. Values are given as median (range) and *n* [%]. LSCS = lower segment caesarean section.

	Monitored group (*n* = 32)	Unmonitored group (*n* = 36)
No. of capillary specimens performed*	118 (0–520)	0 (0)
Frequency of insulin use†	6 [19]	0 [0]
No. of antenatal visits	17 (2–28)	14 (6–33)
No. of hospital admissions	1 (0–6)	0 (0–8)
Delivery type		
Vaginal	22 [69]	25 [69]
LSCS	10 [31]	11 [31]

*$P < 0.005$; †$P = 0.008$.

Secondly, there is evidence that the unmonitored group had more severely abnormal glucose metabolism than the monitored group. Antenatally, the two-hour plasma glucose concentrations were higher in the unmonitored group, and postnatally there was a trend towards more women being left with abnormal glucose metabolism in this group than in the monitored group. Moreover, there does appear to be a trend to more special care admissions and more neonatal hypoglycaemia in the unmonitored group. This difference did not achieve statistical significance, although this may be attributable to a small sample size. The trend could be used by proponents of intensive monitoring to justify the continuation of this practice which otherwise lacks evidence. However, the more severe abnormality of glucose metabolism in the unmonitored group is the more likely explanation for this difference.

All women received dietary advise in this study. This was because of concerns expressed during the development of the protocol about the advisability of not offering any treatment when an abnormality of glucose metabolism had been diagnosed. Both groups received the same basic advice but no comment can be made as to its therapeutic effect. However, a systematic review of this intervention suggests dietary advice to be of little practical benefit[13]. The ACHOIS study will help to address this issue, since women in the untreated arm are not informed of their diagnosis nor given any dietary advice.

Postnatal measurements were incomplete because patients failed to attend for repeat glucose tolerance tests, in spite of several appointments being forwarded. This suggests that it is unrealistic to justify screening for abnormal glucose tolerance in pregnancy on the basis that lifestyle changes, which minimise the risk of diabetes developing later in life, can be made after pregnancy.

Two women in this study developed diabetes mellitus; both were diagnosed postnatally in the unmonitored group. In both cases glycosylated haemoglobin levels were elevated antenatally. One woman was delivered spontaneously at 34 weeks of gestation; the birthweight ratio was 1·18 and, although the infant required admission to a special care baby unit and treatment for hypoglycaemia, this was as likely to be a consequence of prematurity as a diabetic complication. The other woman was delivered spontaneously at 36 weeks of gestation, the birthweight ratio was 1·18 and the infant suffered no untoward consequences. Although these women and their babies all did well, it could be argued that a periodic review of glucose control in the unmonitored group may have been appropriate. This could either have been as a routine part of the protocol or with a third party highlighting any result reaching a threshold value. However, it remains unclear whether such review would have improved outcome and it would need to be assessed as part of a randomised controlled study before becoming widespread practice.

Women in the monitored group of this study suffered significant inconvenience and probably morbidity because of the intensified management regime. They performed a median of 118 capillary samples for glucose estimation and in some cases as many as 500 samples during their pregnancy. In addition, 19% of the monitored group were commenced on insulin with no evidence of benefit to either themselves or their offspring.

The World Health Organisation consensus[10] on the diagnostic criteria for gestational diabetes mellitus, by

Table 6 Postnatal glucose metabolism data. Values are given as median (range) and *n* [%]. IGT = impaired glucose tolerance.

	Monitored group (*n* = 28)	Unmonitored group (*n* = 28)
Fasting glucose (mmol/L)	4·5 (2·7–5·9)	4·4 (2·4–8·8)
Two-hour glucose (mmol/L)	5·1 (2·1–8·5)	5·5 (3·0–13·7)
No. postnatal diabetes	0 [0]	2 [7]
No. postnatal IGT	2 [7]	3 [11]

classifying impaired glucose tolerance and diabetes arising in pregnancy together, will not lead to improvement in the management of women with this condition. Many women with mild impairment of glucose metabolism are, and will continue to be, exposed to intensive intervention, because the outcome of their pregnancy and interventions to alter it are extrapolated from frankly diabetic woman who become pregnant. Indeed, when designing our study we could not agree to reduce the unmonitored group to normal pregnancy management, as members of the multidisciplinary team felt that this would be unsafe.

There is much effort directed at trying to determine the best way to screen for gestational diabetes, but there is still no evidence that there is an intervention available which improves outcome in this condition. We are no further forward than a decade ago, when the conclusion drawn on reviewing all evidence available was that 'except for research purposes, all forms of glucose tolerance testing should be stopped'[14].

CONCLUSION

This study provides no support for the widespread use of intensified management regimens for women with mild abnormalities of glucose metabolism in pregnancy. However, it does support the need for a large, multicentre, randomised trial of such regimens with the power to compare adverse neonatal outcomes as well as maternal inconvenience and morbidity. This study demonstrates the feasibility of such a trial and that not only would such a trial be ethical, but that to continue to treat women with mild abnormalities of glucose metabolism in pregnancy without undertaking such a trial is not evidence based.

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